

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Dynamic Website Interactions Using JavaScript

2. AIM

To construct an interactive, dynamic website using JavaScript DOM manipulation and event listeners.

3. OBJECTIVE

The objectives of this experiment are:

- Capture user interactions via event listeners.
 - Manipulate CSS styles and classes dynamically via JS.
 - Update the document object model in real-time.
-

4. THEORY

Dynamic websites respond to user inputs without requiring full page reloads. JavaScript Event Listeners wait for interactions like clicks or keystrokes, triggering functions that alter the DOM and user interface.

5. ALGORITHM / PROCEDURE

1. Design a styled UI with interactive elements.
 2. Attach event listeners to buttons.
 3. Implement JS functions that modify CSS and HTML upon event trigger.
 4. Test interactions in browser.
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6. PROGRAM

Source File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dynamic Website Using JavaScript</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:linear-gradient(135deg,#74ebd5,#ACB6E5);
}

/* Header */
header{
    background:#333;
    color:white;
    text-align:center;
    padding:20px;
    text-shadow:2px 2px 5px black;
}

/* Navigation */
nav{
    background:#444;
    padding:15px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
    margin:15px;
    font-weight:bold;
}

nav a:hover{
    color:yellow;
}

.des{
    text-align:center;
    padding:50px;
}

.des h2{
```

```

        margin-bottom:20px;
    }

    button{
        padding:10px 20px;
        margin:10px;
        border:none;
        border-radius:8px;
        background:rgb(190,120,250);
        color:white;
        cursor:pointer;
        font-size:16px;
    }

    button:hover{
        background:rgb(130,210,250);
    }

    /* Cards */
    .container{
        display:flex;
        justify-content:center;
        gap:20px;
        padding:30px;
    }

    .card{
        width:250px;
        background:white;
        padding:20px;
        border-radius:10px;
        box-shadow:0 0 10px gray;
        text-align:center;
    }

    .card:hover{
        transform:scale(1.05);
    }

    /* Footer */
    footer{
        background:#333;
        color:white;
        text-align:center;
        padding:15px;
        margin-top:30px;
    }
</style>
</head>

<body>

<header>
    <h1>MY Dynamic Website</h1>

```

```

</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Courses</a>
  <a href="#">Contact</a>
</nav>

<section class="des">

  <h2 id="message">Welcome to My Website</h2>

  <button id="msgBtn">Change Message</button>

  <button id="colorBtn">Change Background</button>

</section>

<div class="container">

  <div class="card">

    <h3>AI</h3>

    <p id="ai">
      AI is useful for the next generation.
    </p>

    <button id="btnAI">
      KNOW MORE
    </button>

  </div>

  <div class="card">

    <h3>EDITING</h3>

    <p id="edit">
      Editing helps you earn money.
    </p>

    <button id="btnEdit">
      KNOW MORE
    </button>

  </div>

  <div class="card">

    <h3>GAME DEVELOPMENT</h3>

    <p id="game">

```

```

        Game Development is a growing career.
    </p>

    <button id="btnGame">
        KNOW MORE
    </button>

</div>

</div>

<footer>

    <p>&copy; A.ROHIT - Dynamic JavaScript Website</p>

</footer>

<script>

document.getElementById("msgBtn").addEventListener("click",changeMessage);

document.getElementById("colorBtn").addEventListener("click",changeColor);


let ai=new AdvancedCourse(
    "AI",
    "Artificial Intelligence enables machines to learn and solve problems.",
    "ai",
    "Advanced"
);

let editing=new Course(
    "Editing",
    "Video editing is useful for YouTube, Films and Social Media.",
    "edit"
);

let game=new AdvancedCourse(
    "Game Development",
    "Game Development uses HTML, CSS, JavaScript, Unity and C#.",
    "game",
    "Intermediate"
);

document.getElementById("btnAI").addEventListener("click",function()
{
    ai.showInfo();

```

```
});

document.getElementById("btnEdit").addEventListener("click",function()
{
    editing.showInfo();
});

document.getElementById("btnGame").addEventListener("click",function()
{
    game.showInfo();
});
function changeMessage()
{
    let name=prompt("Enter your Name");

    document.getElementById("message").innerHTML=
    "Welcome "+name+"!";
}

function changeColor()
{
    let colors=
    [
        "#FFDAB9",
        "#90EE90",
        "#87CEEB",
        "#E6E6FA",
        "#F4A460"
    ];

    let randomColor=
    colors[Math.floor(Math.random()*colors.length)];

    document.body.style.backgroundColor=randomColor;
}

</script>

</body>
</html>
```

7. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Action	Expected Result	Status
TC-1	Trigger interactive events	UI updates dynamically without reloading	Pass

8. OUTPUT

Output 1

JavaScript Arrays & Functions Demonstration

Original Marks: 85, 92, 78, 64, 95, 88

Average Marks: 83.67

Top Scores (≥ 80): 85, 92, 95, 88

Highest: 95 | **Lowest:** 64

9. RESULT

Thus, the dynamic website interactions practice was successfully completed.